

■ 2.3.3 An Example: Defining Your Own Integration Function

Now that we have introduced most of the basic features of patterns in *Mathematica*, we can use them to give a more or less complete example. We will show how you could define your own simple integration function in *Mathematica*.

From a mathematical point of view, the integration function is defined by a sequence of mathematical relations. By setting up transformation rules for patterns, you can implement these mathematical relations quite directly in *Mathematica*.

<i>mathematical form</i>	<i>Mathematica definition</i>
$\int (y + z) dx = \int y dx + \int z dx$	<code>integrate[y_ + z_, x_] := integrate[y, x] + integrate[z, x]</code>
$\int c y dx = c \int y dx$ (c independent of x)	<code>integrate[c_ y_, x_] := c integrate[y, x] /; FreeQ[c, x]</code>
$\int c dx = c x$	<code>integrate[c_, x_] := c x /; FreeQ[c, x]</code>
$\int x^n dx = \frac{x^{n+1}}{n+1}, n \neq -1$	<code>integrate[x_^n_, x_] := x^(n+1)/(n+1) /; FreeQ[n, x] && n != -1</code>
$\int \frac{1}{ax+b} dx = \frac{\log(ax+b)}{a}$	<code>integrate[1/(a_. x_ + b_.), x_] := Log[a x + b]/a /; FreeQ[{a,b}, x]</code>
$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b}$	<code>integrate[Exp[a_. x_ + b_.], x_] := Exp[a x + b]/a /; FreeQ[{a,b}, x]</code>

Definitions for an integration function.

This implements the linearity relation for integrals: $\int (y + z) dx = \int y dx + \int z dx$.

```
In[1]:= integrate[y_ + z_, x_] :=  
integrate[y, x] + integrate[z, x]
```

The associativity of Plus makes the linearity relation work with any number of terms in the sum.

```
In[2]:= integrate[a x + b x^2 + 3, x]  
Out[2]= integrate[3, x] + integrate[a x, x] +  
integrate[b x^2, x]
```

This makes `integrate` pull out factors that are independent of the integration variable x .

```
In[3]:= integrate[c_ y_, x_] :=  
c integrate[y, x] /; FreeQ[c, x]
```

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Mathematica tests each term in each product to see whether it satisfies the `FreeQ` condition, and so can be pulled out.

```
In[4]:= integrate[a x + b x^2 + 3, x]
Out[4]= integrate[3, x] + a integrate[x, x] +
        b integrate[x^2, x]
```

This gives the integral $\int c dx = cx$ of a constant.

```
In[5]:= integrate[c_, x_] := c x /; FreeQ[c, x]
```

Now the constant term in the sum can be integrated.

```
In[6]:= integrate[a x + b x^2 + 3, x]
Out[6]= 3 x + a integrate[x, x] + b integrate[x^2, x]
```

This gives the standard formula for the integral of x^n . By using the pattern `x_<sup>n_</sup>`, rather than `x_<sup>n_</sup>`, we include the case of $x^1 = x$.

```
In[7]:= integrate[x_<sup>n_</sup>, x_] :=
        x^(n+1)/(n+1) /; FreeQ[n, x] && n != -1
```

Now this integral can be done completely.

```
In[8]:= integrate[a x + b x^2 + 3, x]
Out[8]= 3 x +  $\frac{a x^2}{2}$  +  $\frac{b x^3}{3}$ 
```

Of course, the built-in integration function `Integrate` (with a capital I) could have done the integral anyway.

```
In[9]:= Integrate[a x + b x^2 + 3, x]
Out[9]= 3 x +  $\frac{a x^2}{2}$  +  $\frac{b x^3}{3}$ 
```

Here are all the rules you have given for your `integrate` function so far.

```
In[10]:= ?integrate
integrate
integrate/:
integrate[(y_) + (z_), x_] :=
integrate[y, x] + integrate[z, x]
integrate/:
integrate[(c_) (y_), x_] :=
c integrate[y, x] /; FreeQ[c, x]
integrate/: integrate[c_, x_] := c x /; FreeQ[c, x]
integrate/:
integrate[(x_)^n_, x_] :=
 $\frac{x^{n+1}}{n+1}$  /; FreeQ[n, x] && n != -1
```

Here is the rule for integrating the reciprocal of a linear function. The pattern $a_. x_ + b_.$ stands for any linear function of x ; the optional argument b has default value 0; a has default 1. The condition $\text{FreeQ}[\{a,b\}, x]$ checks that both a and b are independent of x .

```
In[10]:= integrate[1/(a_. x_ + b_.), x_] :=
          Log[a x + b]/a /; FreeQ[{a,b}, x]
```

Here both a and b take on their default values.

```
In[11]:= integrate[1/x, x]
Out[11]= Log[x]
```

Here is a more complicated case. The symbol a now matches $2 p$.

```
In[12]:= integrate[1/(2 p x - 1), x]
Out[12]=  $\frac{\text{Log}[-1 + 2 p x]}{2 p}$ 
```

You can go on and add many more rules for integration. Here is a rule for integrating exponentials.

```
In[13]:= integrate[Exp[a_. x_ + b_.], x_] :=
          Exp[a x + b]/a /; FreeQ[{a,b}, x]
```